Marcelo Cambraia Villela

Full Stack Developer .NET | C# | Angular | React

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SUMMARY

Full Stack Developer specialized in C# / .NET, with hands-on experience in CQRS/MediatR, event-driven asynchronous integrations with RabbitMQ (including reliability patterns such as Transactional Outbox), relational databases (SQL Server daily, plus PostgreSQL) and front-end development with Angular/NgRx and React. Builds and maintains a mission-critical web platform with over one million users, handling real-world trade-offs in architecture, transactions and concurrency. Applies SOLID, Clean Code and automated testing with xUnit and Cypress.

PROFESSIONAL EXPERIENCE

Full Stack Developer
Office of the Comptroller General of the Federal District (CGDF) · Brasília, Brazil · Mar 2025 – Present

- Builds and maintains **Participa DF**, a mission-critical government platform with over one million registered users, using C# .NET (Web API) with a **CQRS/MediatR** architecture, NHibernate, SQL Server and an Angular front end.
- Designed and implemented from scratch an **event-driven asynchronous integration via RabbitMQ** (publish/subscribe) between two government systems, with dedicated queues, payload contracts and a dead-letter/retry strategy; evaluated the **Transactional Outbox** pattern to ensure consistency between data persistence and event publishing.
- Fixed a subtle **idempotency** issue caused by NHibernate auto-flush, and implemented transactional locking to restrict operations on an entity while awaiting an asynchronous response from an external system, preserving internal SLAs.
- Identified, through **decompilation**, that a third-party library was silently swallowing BackgroundService exceptions, masking real connectivity failures; validated transactional architecture trade-offs with empirical tests.
- Migrated features from **AngularJS to Angular**, analyzing existing business rules and legacy behavior to preserve compatibility; implemented state management with **NgRx** (Actions, Reducers, Effects, Selectors) for async flows and cross-component state sharing.
- Implemented the integration with **Gov.br** (Brazil's federal single sign-on) using OAuth 2.0 / OpenID Connect, Authorization Code Flow and PKCE; takes part in code reviews through merge requests on GitLab.

PROJECTS

APABB App — Donation and volunteering platform
github.com/MarceloCambraia/apabb-app

- Payment integration with Banco do Brasil APIs: **PIX** (Brazilian instant payments) in production via an mTLS proxy on Google Cloud Run, bank slips (boleto) and BB Pay checkout.
- Full stack in React/TypeScript, serverless back end on Supabase (PostgreSQL, Edge Functions), multi-role Row Level Security, E2E tests with Cypress and LGPD (Brazilian data protection law) compliance.

MetaAssist — AI-powered systematic review and meta-analysis tool
Personal project

- Back end in **ASP.NET Core 8 (Minimal APIs)** with Supabase/PostgreSQL storage and a **React/Vite** front end.
- Integration with **LLMs via OpenRouter** for AI-driven automated study screening.

TECHNICAL SKILLS

Languages: C#, SQL, TypeScript, JavaScript, PHP
Back End & Architecture: .NET, ASP.NET Core, Web API, RESTful APIs, Microservices, CQRS/MediatR, NHibernate, EF Core, LINQ, RabbitMQ (pub/sub, dead-letter/retry, Transactional Outbox)
Databases: SQL Server / T-SQL (daily use), PostgreSQL, Row Level Security, relational modeling
Front End: Angular, NgRx, AngularJS, React, Next.js, Vite, TypeScript, HTML, CSS
Integrations & Payments: Banco do Brasil APIs (PIX, boleto, BB Pay), Gov.br SSO (OAuth 2.0/OIDC/PKCE), mTLS, REST
Cloud & DevOps: Git, GitLab, Jenkins (CI/CD), Docker, Google Cloud Run, Supabase, GitHub
Testing & Quality: xUnit, Cypress (E2E), Postman, Swagger/OpenAPI, SOLID, Clean Code, code review
AI: Claude, GitHub Copilot, Cursor (daily use in development), LLM integration via OpenRouter
Methodologies: Scrum, Kanban, Agile

EDUCATION & LANGUAGES

Associate Degree in Systems Analysis and Development — UniCEUB, Brasília, Brazil — completed 2026
Portuguese: native. English: advanced.